Logan Reed

Ph.D. Student in Mathematics, Temple University

Philadelphia, PA l.reed@temple.edu LoganOReed 0000-0003-0091-9343

Education

Ph.D. **Temple University**, Mathematics — Philadelphia, PA, 2024 – present
- Research interests: numerical linear algebra, Krylov subspace methods, matrix equations, tensor equations, low-rank tensor computations.

M.S. **Rutgers University**, Mathematical Sciences — Camden, NJ, 2023

B.S. **Texas State University**, Applied Mathematics — San Marcos, TX, 2021

Publications

A pipeline for testing drug mechanism of action and combination therapies: From microarray data to simulations via linear-in-flux-expressions: Testing four-drug combinations for tuberculosis treatment — June 2023
Volume 360, 108983.
Christopher Denaro, Nathaniel J. Merrill, Sean T. McQuade, Logan Reed, Karim Azer, Benedetto Piccoli
10.1016/j.mbs.2023.108983 (Mathematical Biosciences)

Talks and Presentations

Nov. 13, 2026: Near-Optimal Convergence of Widlund's Method and Numerical Exploration of Different Implementations. Mid-Atlantic Numerical Analysis Day, Temple University (Philadelphia, PA).

Oct. 24, 2026: Near-Optimal Convergence of Widlund's Method and Numerical Exploration of Different Implementations. Annual Conference of the New York–New Jersey–Pennsylvania section of SIAM, Rutgers University (New Brunswick, NJ).

Attended Workshops

May 4–8, 2026: Asynchronous Methods for Numerical Linear Algebra, ICERM, Brown University (Providence, RI).

Feb. 2–6, 2026: Randomized Numerical Linear Algebra, ICERM, Brown University (Providence, RI).

Experience

Temple University, Graduate Researcher — Philadelphia, PA, 2025 – present, 1 year
Advisor: Daniel Szyld
- Established a near-optimal bound for the convergence of Widlund's method.
- Developed and analyzed different implementations of Widlund's method in the exact and inexact setting.

Computational Mathematics and Modeling Research Lab, Temple University, Graduate Researcher — Philadelphia, PA, 2024
Supervisors: Benjamin Seibold, Gillian Queisser
- Compared different time-stepping schemes for a highly stiff PDE in an asynchronous setting.

Rutgers University, Master's Thesis — Camden, NJ
Advisor: Siqi Fu — 2023

- Developed numerical methods for a constrained shape optimization problem related to the Pólya–Szegő conjecture on polygonal domains.

Applied Mathematics Lab, Rutgers University, Graduate Researcher — Camden, NJ
Supervisor: Benedetto Piccoli — 2022 – 2023 — 1 year

- Contributed to the discretization and implementation of a metabolic pathway model.
- Developed a PDE solver on a large hypergraph representing the metabolic pathway.

Teaching Experience

Spring 2026: Instructor, Precalculus, Temple University.

Fall 2022: Lecturer, College Algebra, Rutgers University.

Fall 2021: Lecturer, College Algebra, Rutgers University.

Fall 2026: Teaching Assistant, Numerical Analysis, Temple University.

Fall 2025: Teaching Assistant, Numerical Analysis, Temple University.

Spring 2022: Teaching Assistant, Calculus I, Rutgers University.

Spring 2022: Mathematics Tutor, Math and Statistics Lab, Rutgers University.

Fall 2021: Mathematics Tutor, Math and Statistics Lab, Rutgers University.

2017–2020: Mathematics Tutor, Math CATs, Texas State University.

Service and Outreach

2025–Present: President, Temple University SIAM Student Chapter.

2025–Present: Graduate Seminar Organizer, Department of Mathematics, Temple University.

Honors and Awards

Distinguished Thesis Award — Rutgers University, May 2023

Mathematical Sciences Scholarship Award — Rutgers University, May 2022

Technical Skills

Languages: MATLAB, Python, C, C++, Fortran

Frameworks: NumPy, SciPy, BLAS, LAPACK

Tools: Linux, Git, Bash, LaTeX